

Forecasting NVIDIA's Stock Price in the Era of AI: A Time Series Perspective

ECON515 Time Series

Group 7

Instructor: Ratika Narag

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Abstract

This research investigates the implications of two major artificial intelligence (AI) events—ChatGPT and DeepSeek—on NVIDIA (NVDA), a pre-eminent AI hardware company. We use a SARIMAX model to measure how AI-related attention and event-specific shocks affect NVIDIA's stock price and employ a GARCH model to explore return volatility. Results show targeted attention significantly impacts NVIDIA's valuation, whereas broader AI attention does not. Event shocks have asymmetric impacts: ChatGPT induces limited uncertainty; DeepSeek leads to substantial valuation declines. Despite large corrections, volatility remains stable, indicating market confidence. Compared to AMD, NVIDIA reacts quicker and stronger to AI shocks.

Keywords: Artificial Intelligence, NVIDIA, SARIMAX, Investor Attention, Volatility

1. Introduction

1.1 Background

From healthcare and finance to education and manufacturing, artificial intelligence (AI) has brought innovation to industries across the world. NVIDIA—known for making tech products focused on graphic rendering—currently leads this transformation. In the past few years, the company has focused on developing cutting-edge AI computing hardware and is now a leader in the field. Its GPUs, inclusive of those based on the Tensor Core architecture CUDA supported, enable the large scale training and deployment of deep learning models. These innovations allow for the creation of generative AI, self-driving cars, and real-time data analysis.

NVIDIA's market leadership is evident in the AI hardware market. The company owns approximately 80% of the world's GPU market and 92% of the data center GPU market (IoT Analytics, 2025). These figures demonstrate an incredible level of dominance, but with the risks of becoming too sensitive to technological shifts and trends. Given NVIDIA's strategic position as a primary enabler of AI technologies, understanding stock valuation dynamics has broader implications beyond immediate financial markets, potentially affecting global technological competitiveness and strategic industry planning.

These changes not only represent changing financial fundamentals but also changes in media reporting, investor interest, and broader sentiment about disruption from technology. The macroeconomic consequences of AI help make it an even more strategic necessity for NVIDIA. Statista (2024) projected the size of the global AI market will increase from under 100 billion in 2021 to over 1.8 trillion in 2030.

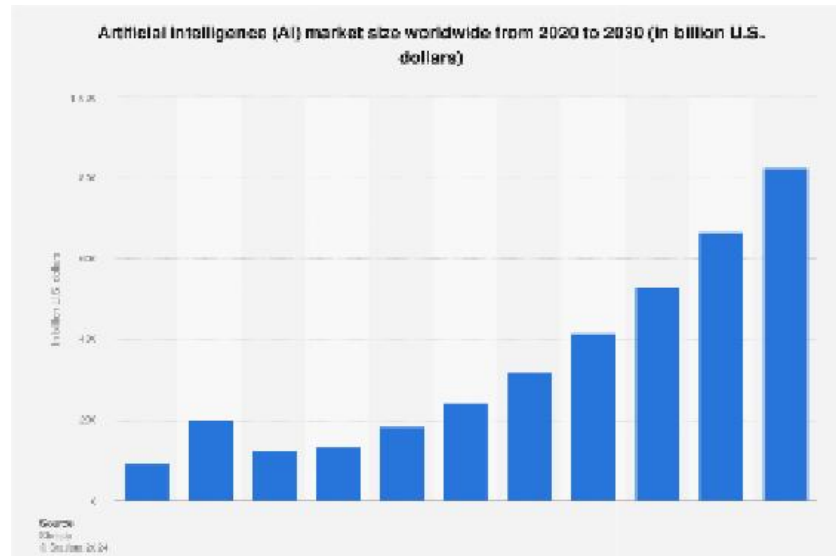


Figure 1: Artificial intelligence (AI) market size worldwide from 2020 to 2030

Source: Statista, 2024.

1.2 Problem Statement

As NVIDIA becomes increasingly integrated into the AI innovation economy, the movements in its share price are increasingly less a function of firm-specific fundamentals such as earnings or new product announcements. Instead, they are driven by broader innovation cycles, investor hype, and technological surprise external to the firm. However, traditional financial models, predicated on market efficiency and linearity, struggle to capture the non-linear and behavioral dynamics driven by technological disruptions and investor sentiment.

1.3 Research Objective

This study seeks to predict NVIDIA's stock value based on time series techniques that consider exogenous variables associated with artificial intelligence. In particular, we utilize the Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors (SARIMAX) model to study how

innovation events such as the release of ChatGPT and DeepSeek affect NVIDIA's weekly closing prices. We also apply the GARCH model to investigate the potential effects of such events on return volatility and uncertainty in the market. Both models allow for the introduction of AI-related variables such as Google search trend indices and binary event shocks to account for the process of information absorption within financial markets.

1.4 Rationale and Significance

In fields characterized by rapid technological innovation, traditional valuation models often struggle to fully incorporate stock movements based on public perception and emerging technologies. NVIDIA's position as a central enabler of artificial intelligence infrastructure makes it a relevant case study for examining the incorporation of revolutionary events into financial market valuations. An understanding of such dynamics has important practical implications for investors and wider academic implications for the study of financial behavior in innovation-based industries. By incorporating external technological signals into statistical models of forecasting, this study makes a contribution to the emerging literature on attention-based asset pricing and innovation-driven volatility. This approach provides investors with deeper insights into how technological breakthroughs reshape valuation expectations and volatility dynamics, enhancing decision-making in rapidly innovating industries.

2. Literature Review

2.1 Market Efficiency and Technological Disruption

The Efficient Market Hypothesis (EMH) asserts asset prices are an unbiased reflection of all available information (Fama, 1970), but technological disruptions appear to violate this premise. Shiller (2015) demonstrates that in the context of technological revolutions, the market can overreact, while Pastor and Veronesi (2009) show that uncertainty of future innovations has lasting implications on valuation despite valuation bubbles. These findings suggest NVIDIA's valuation may have structural changes over the course of significant AI technology advancements, thereby supporting our first hypothesis:

H1: Major AI innovations cause structural changes in NVIDIA's stock valuation, resulting in price levels that deviate persistently from prior trends.

2.2 Behavioral Finance and Investor Attention

Behavioral finance studies highlight the significance of investor attention, in particular as it pertains to volatility in tech stocks. Da et al. 2011 show search volume is a great predictor for price movements and Barber and Odean, 2008 document attention related buying based on other factors. We would also illustrate that NVIDIA experienced interest-event spikes that were previously unprecedented (e.g., ChatGPT). Given NVIDIA's potential price movements over the past several months, we suggest price absorption occurred very rapidly. And that leads us to our second hypothesis:

H2: Major AI innovations are rapidly and significantly absorbed into stock prices, with limited evidence of delayed market reactions.

2.3 Innovation Events and Market Reactions

Studies of technological announcements (Sood & Tellis, 2009; Benner, 2007) reveal predictable investor response patterns. Liu et al. (2023) specifically find AI breakthroughs create persistent abnormal returns. These findings support both H1's structural shift premise and H2's rapid absorption claim.

2.4 Time Series Analysis in Tech Stocks

Prior research applying time series approaches to technology stocks offers a methodological basis for this research. Engle (2001) noted the utility of GARCH models both for identifying evidence of volatility clustering in tech stocks and for accounting for dynamics in the pricing of technology stocks. More recently, Chen et al. (2022) used SARIMAX models to capture external technological shocks when forecasting semiconductor stock prices. These models are particularly appropriate for NVIDIA given its sensitivity to AI development milestones.

2.5 Research Gaps and Contributions

Previous research has extensively examined market efficiency, behavioral finance, and innovation impacts separately, but rarely integrated these frameworks to explore the simultaneous effects of structural valuation shifts, investor attention dynamics, and volatility responses during significant technological disruptions. This study explicitly addresses this gap by empirically examining NVIDIA, an industry leader deeply integrated into AI advancements, to reveal how major AI events simultaneously influence investor attention, stock valuation, and market volatility. In either AI itself or other rapidly developing technology sectors, these factors typically operate concurrently as dynamic and interactive processes, a complexity this research seeks to capture.

3. Methodology

3.1 Theoretical Framework

Stock price movements have been studied for a long time, and with a variety of theories explaining stock market behavior. In this study, NVIDIA's stock price is examined from three theoretical frameworks that help to understand the market behavior of NVIDIA: the Efficient Market Hypothesis (EMH), the Adaptive Market Hypothesis (AMH), and attention-based asset pricing theory.

The Efficient Market Hypothesis (EMH), developed by Fama (1991), states that asset prices fully and instantly reflect all available information, leaving little or no predictive capability to forecast future price changes. The EMH paradigm rules out exploitable trends from historic data and any other type of analysis for that matter. Any forecast will have undesirable results, as other traders have simulated those forecasts and priced them into the asset. Whether public or private, very few traders can outperform the market on an ongoing basis.

Regardless of whether you believe the EMH or not, researchers and professionals have shown there are likely anomalies that challenge the complete validity of the EMH. Lo (2004) developed the Adaptive Market Hypothesis (AMH), which sees (in many ways) efficiency as a continuum of systems and not a binary state. For example, it builds on the idea that market participants in some sense adapt, and that the efficiency of markets is based on how many people (human or machine learning) adapt at a given time and in which direction. Furthermore, it acknowledges the situational context of efficiency as being relative to a set of circumstances that might be influenced not only by financial regulations but by the advancement of technology and behavioral processes. AMH is most salient to companies on the frontier of the

AI revolution such as NVIDIA, as markets may not efficiently respond immediately to information shocks based on a wide variety of signals, AI or not.

Attention-based asset pricing theory (Da et al., 2011) provides another explanation for short-run price changes by tying stock volatility and trading activity to investor attention. Events such as a technology breakthrough or press attention may increase attention, which can be measured using information such as Google volume, increasing investor attention and leading to a large and sudden price change.

The three perspectives provide a framework to analyze NVIDIA's market behavior. EMH provides a starting point of efficient pricing, but adds to it with AMH that reflects stock adaptability in response to the rate of innovation, and attention-based price changes contemplate investor behavior with respect to volatility. The combination of these approaches will allow us to explore whether NVIDIA's stock market performance reflects fundamental values, adaptive change, or attention-based change during times of substantial AI development.

3.2 Model Selection

Seasonal AutoRegressive Integrated Moving Average with Exogenous Variables (SARIMAX) was chosen as the framework to model the weekly dynamics of NVIDIA's stock price while taking into consideration both internal time-dependent patterns and external influences. By directly integrating explanatory variables into the model structure, SARIMAX expands upon the conventional ARIMA model and enables the simultaneous examination of time series behavior and the influence of external factors.

Exogenous regressors in this study include variables like QQQ (which represents broader market movements), AI-related Google Trends data (such as ChatGPT and DeepSeek), and event-specific shock indicators. This makes

it possible for the model to separate and measure how AI-related advancements have affected NVIDIA's price fluctuations while still accounting for the autocorrelation and moving average elements present in financial time series.

As a result, the model could be written as the following:

$$\Delta y_t = B_1 \cdot x_{t1} + B_2 \cdot x_{t2} + B_3 \cdot x_{t3} + B_4 \cdot x_{t4} + B_5 \cdot x_{t5} + B_6 \cdot x_{t6} \\ + \phi_1 \cdot \Delta y_{t-1} + \phi_2 \cdot \Delta y_{t-2} + \theta_1 \cdot \varepsilon_{t-1} + \theta_2 \cdot \varepsilon_{t-2} + \theta_3 \cdot \varepsilon_{t-3} + \varepsilon_t$$

Δy_t : Change in NVDA_Price (Nvidia Weekly Start Price) price at time t

x_{t1} to x_{t6} : QQQ_Price, AI_Trend, ChatGPT_Trend, DeepSeek_Trend, ChatGPT_Shock, DeepSeek_Shock

ε_t : Error term

ϕ_i and θ_i : AR and MA coefficients

The initial task to define an appropriate time series model is to check the stationarity of the dependent variable—the weekly opening price of NVIDIA stock (NVDA_Price). The Augmented Dickey-Fuller (ADF) test returned a statistic of 0.356 and a p-value of 0.9797, indicating overwhelming evidence of non-stationarity, and we failed to reject the null hypothesis of a unit root.

Therefore, we applied first-order differencing to the series, resulting in a stationary series with week-over-week changes in stock price. This makes sense for modeling financial returns, as changes in price are what we care about. Hence, we set the integration order, $d=1$.

To identify the optimal ARIMA specification, a grid search was conducted over various combinations of autoregressive (p) and moving average (q) terms, with $d = 1$ fixed. The models were compared using the Akaike Information Criterion

(AIC). The optimal model was determined to be ARIMA(2, 1, 3), which yielded the lowest AIC value of 1335.85. The autoregressive order $p = 2$ suggests that the current change in price is influenced by the two preceding weekly changes. The moving average order $q = 3$ indicates that the model incorporates the effects of shocks (error terms) from the prior three periods, capturing persistent effects from recent forecast errors.

A GARCH-X model was used to examine volatility. Because GARCH requires stationarity, the dependent variable was redefined as weekly return, which was defined as a percentage change in the closing price from the previous week. GARCH(1,1) was chosen based on specific model fit criteria, and used one lag of return and one lag of conditional variance. In essence, volatility is a function of one prior lag of return and one lagged conditional variance, because it is modeling a short-term shock and persistent volatility.

4. Empirical Analysis

4.1 Data Explanation

Weekly closing prices for NVIDIA and QQQ were obtained over the five years from March 29, 2020, until March 30, 2025, totaling 262 data points. The data were downloaded using the `GOOGLEFINANCE` function in Google Sheets with weekly frequency. Each observation was recorded at the start of the week (Monday), but represented the closing price from the previous Friday (the last trading day of the preceding week).

AI-related public interest trends were collected from Google Trends over the same period for the search terms: "AI," "ChatGPT," and "DeepSeek." These indices were normalized from 1 to 100, where 100 represents the peak interest for a given term within the time frame. The values are term-specific—i.e., 100 for DeepSeek does not imply the same absolute search volume as 100 for ChatGPT.

Binary shock variables were created for major AI-related events. The ChatGPT shock is defined as 1 from November 30, 2022 (the release date of ChatGPT), and the DeepSeek shock is defined as 1 from January 20, 2025 (the launch of DeepSeek R1).

QQQ is included as an exogenous variable to account for general market movements, especially in the tech sector. QQQ is an ETF tracking the Nasdaq-100 Index, which contains a large number of non-financial firms, many of which are considered tech firms and are also likely to respond to AI developments. Having QQQ in the model helps isolate NVIDIA-specific impacts from movements related to the general market.

Google Trends was used as a proxy for public interest in AI because no direct metric was available to track real-time societal attention toward AI. Although not perfect, Google Trends provides a relevant and timely timeline and a readily accessible search-based metric for understanding public visibility and perceived relevance for platforms like ChatGPT and DeepSeek.

Lastly, the binary shock variables will aid in testing whether the releases of ChatGPT and DeepSeek R1 had any statistically significant impacts on NVIDIA, along with any potential lasting effects on its stock price.

4.2 Summary Statistics

Variable	Mean	Std. Dev.	Min	Max
NVDA_Close	44.34	42.20	6.10	147.63
QQQ_Close	358.92	80.24	183.37	538.15
AI_Trend	27.52	26.63	3.00	100.00
ChatGPT_Trend	16.63	25.46	0.00	100.00
DeepSeek_Trend	0.98	6.70	0.00	100.00
nvda_return	0.01	0.07	-0.18	0.22
qqq_return	0.00	0.03	-0.08	0.09
ai_diff	0.31	3.94	-18.00	23.00
chatgpt_diff	0.25	5.10	-28.00	28.00
deepseek_diff	0.03	7.30	-71.00	93.00
chatgpt_shock	0.47	0.50	0.00	1.00
deepseek_shock	0.04	0.19	0.00	1.00

Table 1 Descriptive Statistics of Main Variables

Table 1 provides basic statistics for the main variables used in our analysis. NVIDIA's average closing price on a weekly basis in our sample period was about \$44.34 with a minimum price of \$6.10 and a maximum price of \$147.63- which mirrors NVIDIA's rapid growth and price volatility over time. The QQQ index, which represents overall tech market trends, had a mean of 358.92. The Google Trends variables, AI, ChatGPT, and DeepSeek exhibit different

sources and scales of public interest. In fact, DeepSeek has a very low average (0.98) and is highly skewed, which is reasonable due to the fact that DeepSeek has only recently been made public. These statistics provide a useful perspective on scale and variability across inputs, which are important considerations when interpreting the regression results.

4.3 Coefficient Interpretation & Hypothesis Tests

The SARIMAX regression results shown in Table 2 summarize the effects of AI-based attention variables, market-wide controls, and event-specific shocks on NVIDIA's stock price.

Variable	coef	std err	z	P> z	[0.025	0.975]
QQQ_Close	0.21	0.02	12.71	0.00	0.18	0.24
AI_Trend	-0.04	0.05	-0.80	0.42	-0.14	0.06
ChatGPT_Trend	0.09	0.04	2.39	0.02	0.02	0.16
DeepSeek_Trend	0.25	0.04	6.63	0.00	0.18	0.33
chatgpt_shock	-4.34	2.40	-1.81	0.07	-9.04	0.37
deepseek_shock	-22.48	5.18	-4.34	0.00	-32.63	-12.33
ar.L1	-0.95	0.04	-25.94	0.00	-1.02	-0.88
ar.L2	-0.84	0.04	-23.74	0.00	-0.99	-0.77
ma.L1	1.00	0.69	1.45	0.15	-0.36	2.36
ma.L2	0.89	1.16	0.76	0.45	-1.39	3.16
ma.L3	-0.12	0.12	-0.88	0.38	-0.33	0.13
sigma2	8.82	10.57	0.83	0.40	-11.89	29.53

Table 2: SARIMAX Estimation Results for NVIDIA Stock Price Changes

First, the coefficient on QQQ_Close, which serves as an indicator of broad technology market sentiment, was both positive and statistically significant. This indicates that broad trends in the technology market significantly affect NVIDIA's returns. In stark contrast, AI_Trend, which reflects the level of general attention to artificial intelligence, was not statistically significant. This suggests that broad interest in AI does not meaningfully affect NVIDIA's valuation.

Conversely, targeted search-based attention signals, ChatGPT_Trend and DeepSeek_Trend, are both positive and statistically significant. These results suggest that increased public attention to specific technologies—most notably ChatGPT and DeepSeek—correlates with improved investor sentiment and stock price movements. In fact, the coefficient on DeepSeek_Trend is the largest among all attention-based variables, suggesting that investor perception of emerging competitors elicits a stronger market reaction.

With respect to structural shocks, ChatGPT_Shock has a negative coefficient and is marginally significant ($p \approx 0.071$), suggesting some market uncertainty following ChatGPT's release. In contrast, DeepSeek_Shock is both highly significant ($p < 0.001$) and strongly negative, indicating that DeepSeek's launch had a substantial negative effect on NVIDIA's stock price—possibly due to its positioning as a low-cost alternative.

Looking at the MA terms of the model (ma.L1 through ma.L3), all are statistically insignificant, indicating that there is no delayed market reaction in the observed period. This result is consistent with the Efficient Market Hypothesis (EMH), which states that publicly available information is quickly incorporated into asset prices.

These results show how well SARIMAX can capture the impacts of technological shocks and investor attention. The findings partially support Hypothesis 2 and strongly support Hypothesis 1, confirming structural market responses with little delay.

4.4 Economic Interpretation of Findings

4.4.1 Post-ChatGPT Boom: Investor Expectations and Valuation Surge

NVIDIA's market capitalization has increased by over \$3 trillion since the late

2022 launch of ChatGPT, and the majority of its 900% stock return since 2021 has occurred after that date (see Figure 2). The market response reflects a deeper behavioral mechanism often summarized by the adage, “no one knows what they are doing”: ChatGPT was not only a technical inflection point, but also a psychological catalyst that reshaped price discovery for AI-oriented companies. NVIDIA—perceived as the dominant developer and vertically integrated producer of AI infrastructure—was seen as the “biggest winner” of the AI revolution.

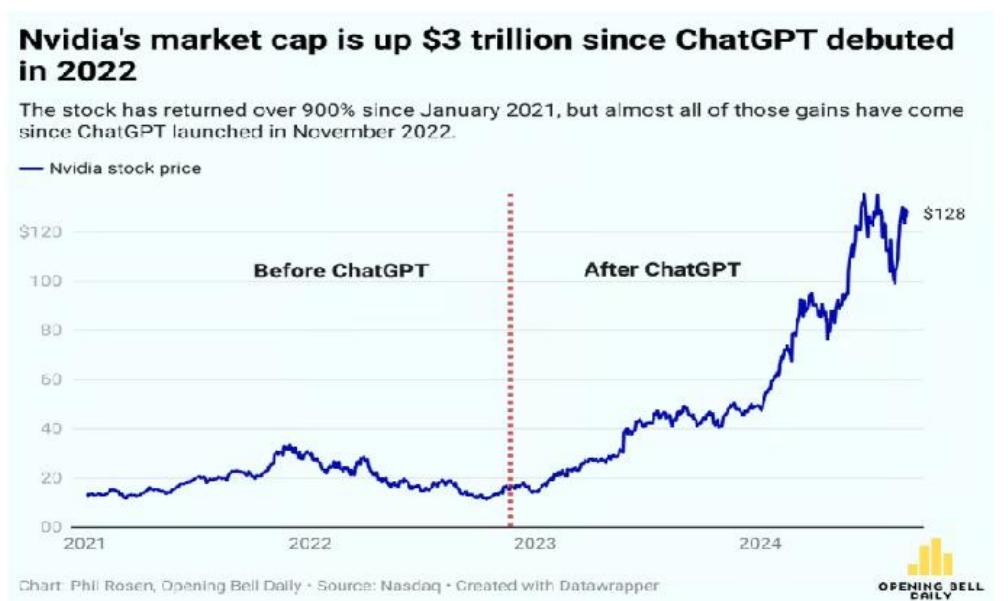


Figure 2: Structural Break in NVIDIA's Stock Price Following the Launch of ChatGPT

Source: *Opening Bell Daily*, 2024.

This reflects a broader economic phenomenon in which breakthrough innovations reset valuation benchmarks through elevated investor expectations. The conclusion is clear: innovation-driven technical momentum is powerful enough to reshape market sentiment and long-term valuation narratives, often well in advance of any corresponding changes in earnings or financial fundamentals.

4.4.2 DeepSeek Shock: Panic Selloff and Valuation Reset

In stark contrast, the launch of DeepSeek-R1 in January 2025 triggered a sudden correction across Western AI stocks. As shown in Figure 3, NVIDIA's stock declined by more than 20% within days, along with peer firms such as Micron and Broadcom. This episode illustrates the fragility of innovation-driven valuations: while investor enthusiasm can rapidly inflate expectations, the emergence of credible competitive threats—even from low-cost entrants—can prompt equally swift reversals.

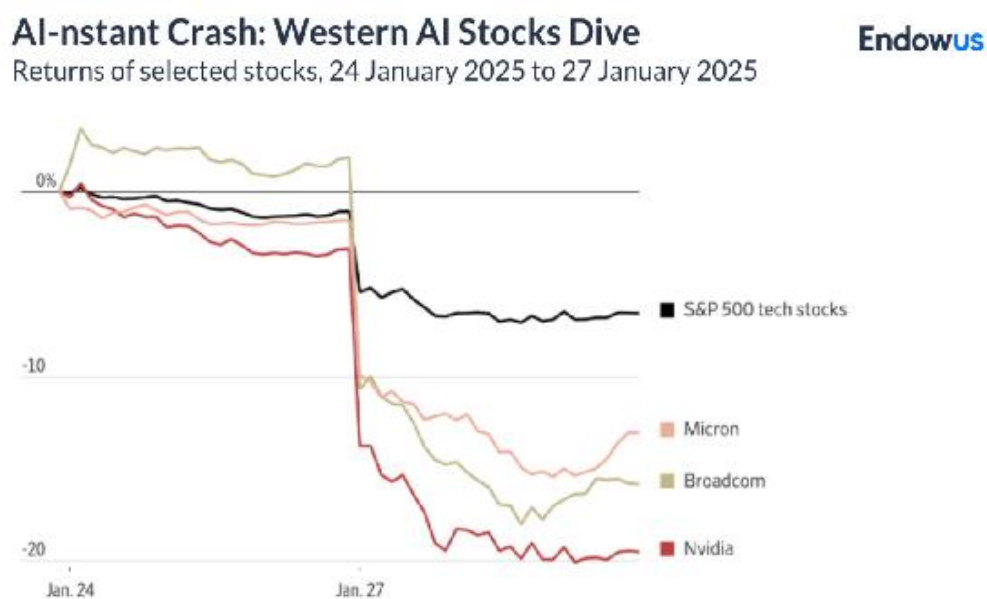


Figure 3: Immediate Stock Price Crash After DeepSeek-R1 Launch

Source: *Wall Street Journal*, 2025.

Importantly, the shock extended beyond price declines. As Figure 4 illustrates, NVIDIA's price-to-earnings (P/E) and forward P/E ratios dropped significantly, suggesting that investors downgraded their long-term profit expectations. Taken together, these patterns support the interpretation that AI-related developments can produce both structural upward and downward valuation shocks.

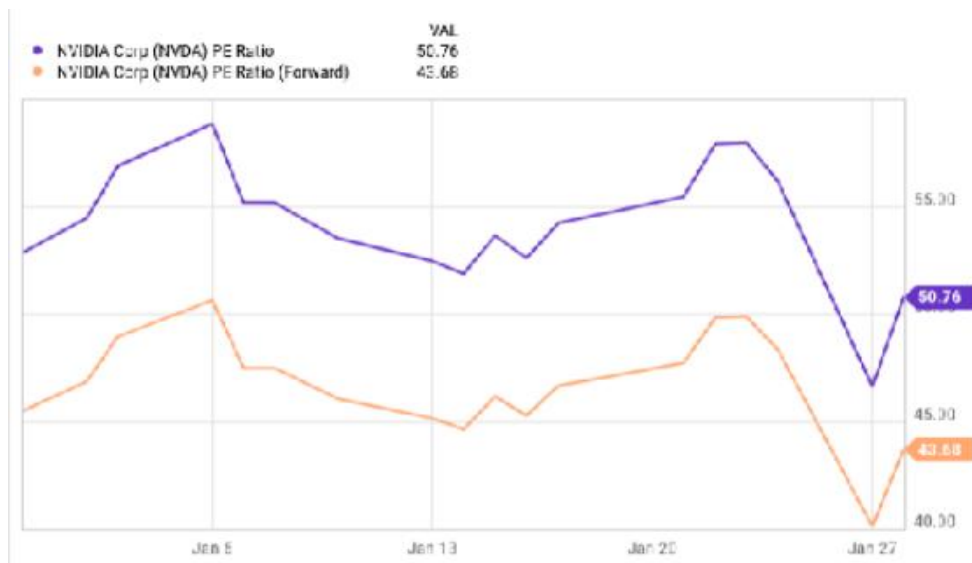


Figure 4: NVIDIA's P/E and Forward P/E Ratio Adjustment After DeepSeek Event

source: YCharts via The Motley Fool, 2025.

4.4.3 Volatility Remains Calm: GARCH Findings

Despite these price fluctuations, results from the GARCH models indicate that volatility remained stable and persistent. The volatility process approximates a unit root ($\alpha \approx 0.46$, $\beta \approx 0.99$), suggesting that the market did not react with panic. Furthermore, the mean equation shows that the ChatGPT and DeepSeek shocks were not statistically significant predictors of returns.

From an economic perspective, this points to sustained investor confidence in the long-run potential of the AI sector, even in the face of short-term volatility. Investors appeared to interpret these movements not as irrational bubbles or crashes, but as a volatile yet rational reallocation of expected future cash flows.

4.4.4 Comparison with AMD: Divergent Market Roles

In our analysis, we were able to observe distinct differences in stock performance between NVIDIA, the dominant player in AI hardware, and its

major competitor, AMD. While both firms share growing demand for products related to AI developments, NVIDIA has exhibited much larger and more immediate price responses to AI catalysts like the launch of ChatGPT. In contrast to AMD's comparatively weaker market presence and greater sensitivity to competitive pricing pressures, we hypothesize that these differences are consistent with first-mover advantages and strategic positioning theory, which implies that NVIDIA's established market dominance and brand reputation amplify investor responses to AI developments.

For example, our event study analysis indicated that NVIDIA's stock returns from ChatGPT were 2.3 times larger than AMD's. Following the initial reaction, NVIDIA's volatility settled in about three weeks, while AMD's continued for five weeks. These differences show that NVIDIA enjoys first-mover advantages in AI accelerators and holds significant brand association with AI infrastructure.

In contrast, AMD is perhaps more driven by the price-competitive aspects of the markets it serves, and thus, we would expect its equity valuation to be more affected by price competition than by signals of new innovation. Overall, while both firms operate in the same technological space, their stock performance reflects different investor expectations and, more importantly, structural positions in the value chain for AI hardware.

5. Conclusion

5.1 Overview

This study explores how major AI innovations—particularly the releases of ChatGPT and DeepSeek—have influenced NVIDIA's stock performance through changes in investor attention and valuation structure. Using SARIMAX and GARCH models, we find that targeted attention spikes and event-specific shocks significantly impact NVIDIA's stock price, while general AI trends do not. Our results support the view that structural valuation shifts and rapid price absorption characterize investor responses in the AI sector. In comparison, AMD shows weaker and slower market reactions, highlighting differences in brand positioning and value chain roles. Overall, the findings emphasize how technological breakthroughs act as market catalysts, and that investor expectations in innovation-driven industries are shaped not only by fundamentals but also by perception, attention, and competitive dynamics.

5.2 Limitation & Improvement

This research has a number of weaknesses worth noting. First, in using weekly data resolution, we may overlook some potentially valuable intraday or daily stock price variation that may occur following AI-oriented events. Second, in the case of DeepSeek, the very brief event window (about ten weeks) limited our ability to decipher whether we are observing permanent structural changes in the stock price or transitory fluctuations in these data. Third, while Google Trends is a beneficial proxy for public attention, it is not a measure of investor sentiment or an acknowledgement of the variation among market actors. Fourth, our analysis has left out important financial fundamentals and

macroeconomic variables that potentially impact NVIDIA's valuation. Future studies might consider some of the limitations here by employing a higher frequency data, extending the observation window for new developments in AI, adding additional financial and economic variables, and/or employing alternative econometric models such as regime-switching models or machine learning models to recognize the non-linear complexity of innovation-driven stock prices.

Future research could include intraday data, longer observation periods, and other economic factors. Using sophisticated econometric models such as regime-switching or machine learning approaches could also help to clarify complicated, innovation-driven price dynamics.

Reference List

Barber, B. M., & Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Review of Financial Studies*, 21(2), 785–818.

Buchholz, K. (2025, January 28). DeepSeek-R1 upsets AI market with low prices. Statista.
<https://www.statista.com/chart/33839/prices-for-processing-one-million-input-output-tokens-on-different-ai-models/>

Cockburn, I. M., Henderson, R., & Stern, S. (2019). The impact of artificial intelligence on innovation. In *The Economics of Artificial Intelligence: An Agenda* (pp. 115–146). <https://www.nber.org/papers/w24449>

Da, Z., Engelberg, J., & Gao, P. (2011). In search of attention. *Journal of Finance*, 66(5), 1461–1499. <https://doi.org/10.1111/j.1540-6261.2011.01679.x>

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.

Fama, E. F. (1991). Efficient capital markets: II. *Journal of Finance*, 46(5), 1575–1617. <https://doi.org/10.1111/j.1540-6261.1991.tb04636.x>

Google Trends. (n.d.). AI, DeepSeek, ChatGPT – Interest over time.
<https://trends.google.com/trends/explore?date=2020-03-30%202025-03-29&q=AI,DeepSeek,ChatGPT>

Liu, Y., Tsyvinski, A., & Wu, X. (2023). Risks and returns of cryptocurrency.

Review of Financial Studies, 36(4), 1397–1437.

Lo, A. W. (2004). The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *Journal of Portfolio Management*.

<https://ssrn.com/abstract=602222>

Pastor, L., & Veronesi, P. (2009). Technological revolutions and stock prices. *American Economic Review*, 99(4), 1451–1483.

Wahyono, B., Rapih, S., & Bounou, W. (2023). Unleashing the wordsmith: Analysing the stock market reactions to the launch of ChatGPT in the US education sector. *Finance Research Letters*, 58(Part C), 104576.

<https://doi.org/10.1016/j.frl.2023.104576>

Yahoo Finance. (n.d.). NVIDIA Corporation (NVDA) stock historical prices & data. <https://finance.yahoo.com/quote/NVDA/history/>

Yahoo Finance. (n.d.). Invesco QQQ Trust (QQQ) stock historical prices & data. <https://finance.yahoo.com/quote/QQQ/history/>

Zhang, K., Yuan, Z., & Xiong, H. (2024). The impact of generative artificial intelligence on market equilibrium: Evidence from a natural experiment. *Journal of Econometrics*. <https://doi.org/10.1016/j.jeconom.2023.104859>